

# HARSH SINGH

AI / ML & GENERATIVE AI ENGINEER

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AI/ML Engineer focused on Generative AI, NLP and LLM-based systems, with hands-on experience building RAG pipelines, LangGraph agents and prompt-injection guardrails using Python, PyTorch and HuggingFace Transformers. Skilled in SQL and ETL/data pipelines, and familiar with AWS, Azure, Docker, Git and GitHub Actions CI/CD for packaging and shipping ML work. Builds with reliability and evaluation in mind (automated tests, bounded retries, grounded citations, red-team testing) and translates complex technical work into clear outcomes for cross-functional stakeholders — comfortable owning issues rather than just flagging them.

## EDUCATION

**[DTU] Delhi Technological University**

(Formerly Delhi College of Engineering)

B.Tech, Electrical Engineering | 2022 – 2026

## ONLINE COURSEWORK

- [Machine Learning Specialization](#)
- [Deep Learning Specialization](#)
- [Generative AI with Large Language Models](#)
- [Generative AI for Everyone](#)
- [Hugging Face Transformers & LLM Course](#)

## TECHNICAL SKILLS

**Programming & Libraries:** Python, SQL, Git

**Big Data:** PySpark (fundamentals)

**ML Algorithms:** Regression, Classification, Clustering (K-Means, PCA), Ensemble Methods (Random Forest, XGBoost), Feature Engineering, Hyperparameter Tuning

**Deep Learning:** CNNs, RNNs/LSTMs, Transformers (Attention, BERT/GPT), Transfer Learning — TensorFlow, PyTorch, Keras, scikit-learn

**Generative AI & LLM Engineering:** NLP, LLMs, RAG, LangChain, LangGraph, FAISS, Vector Databases, Prompt Engineering, LoRA/QLoRA Fine-tuning, LLM Agents, Prompt-Injection Guardrails

**ETL & Data Warehousing:** ETL pipeline design, Data Warehousing, DBMS, EDA, Data Cleaning & Validation

**Cloud & MLOps:** AWS, Azure, Docker, Kubernetes, Git, GitHub Actions, CI/CD, MLOps (version control, automated testing, deployment)

**Visualization:** Tableau, Power BI, Matplotlib, Seaborn

## PROFESSIONAL EXPERIENCE

**Data Science Intern | ExpressEarth**

March 2026 – August 2026

- Extracted and cleaned a customer booking dataset (~8,000 records, 12 attributes including demographics, destination, trip type, spend, and booking channel), preparing it for exploratory analysis and segmentation.
- Performed exploratory data analysis (EDA) to uncover booking behavior patterns across customer demographics, trip types, and destinations, informing the feature set used for segmentation.
- Built customer segments using an RFM (Recency, Frequency, Monetary) framework — combining recency of last booking, booking frequency, and total/average trip spend — to distinguish high-value and low-engagement traveler groups.
- Designed interactive Power BI dashboards to visualize customer segments, booking trends, and spend patterns for the team, translating EDA findings into a reusable reporting view.

**Industrial Trainee — Data Analytics | NTPC, Dadri**

June 2025 – August 2025

- Analyzed operational data using SQL and Python, applying structured data cleaning and exploratory analysis to support electricity-consumption forecasting for a regenerative feed heating system.
- Analyzed real-time SCADA monitoring and data pipelines end-to-end, gaining direct exposure to ETL processes, data quality, reliability, and governance considerations for mission-critical, predictive-maintenance systems.
- Collaborated with a cross-functional trainee team to extract, validate, and document datasets — including writing SQL queries for data preparation — used in the group's electricity consumption prediction model.

## PROJECTS

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### Doorman — Prompt-Injection Guardrails for an AI Recruiting Agent — [GitHub](#) | 2026

*Layered defense for a simulated resume-screening agent*

**Tools:** Python, LLM tool-calling agent

- Built Doorman, a guardrails layer for a simulated recruiting-platform agent that reads candidate resumes and portfolio PDFs, scores them, emails candidates, and writes status to an ATS — defending against prompt injection hidden in untrusted documents.
- Implemented layered defense: input classification, hard isolation of untrusted document content from instructions, per-context tool allowlists, output scanning, and explicit confirmation before irreversible actions such as sending emails.
- Authored a 60-attack red-team suite grouped by attack family, and evaluated attack success rate before and after the guardrails alongside the false-positive rate on 100 benign resumes.
- Added structured logging so every blocked action names the rule that fired, and documented a bypass found during red-teaming and closed it.

### AI Research Agent — [Live App](#) | January 2026

*Stateful, human-in-the-loop research agent*

**Tools:** Python, LangGraph, LangChain, Tavily, Streamlit

- Built a stateful research agent as an 8-node LangGraph StateGraph (planner → review\_plan → researcher → writer → critic → refine → review\_report → finalize) that plans, searches, and synthesizes a cited report on a given topic.
- Implemented two human-in-the-loop checkpoints using LangGraph's interrupt()/Command(resume=...), letting a user approve, edit, or reject both the research plan and the final report before the agent proceeds.
- Built a self-correction loop where a critic node scores each report draft and triggers additional research rounds (bounded by a configurable revision cap, MAX\_REVISIONS) when topic coverage falls below a configurable quality threshold (QUALITY\_THRESHOLD), improving average report completeness scores by ~25% over the uncorrected first draft.
- Grounded every claim in real-time web evidence via the Tavily search API, with inline citations and a verified Sources list assembled directly from agent state rather than generated by the LLM, reducing unsupported claims compared to a no-citation baseline.
- Added exponential-backoff retry logic (tenacity) for search/LLM calls and a fully offline mock mode, backed by a pytest suite covering 15+ test cases that runs the entire graph end-to-end with zero API calls.

### VulgarVeto — [Live App](#) | November 2025

*Audio profanity detection & censorship tool*

**Tools:** Python, OpenAI Whisper, pydub, gTTS, Streamlit

- Built VulgarVeto, a tool that transcribes uploaded audio using OpenAI's Whisper model, flags words against a configurable 18-term profanity list, and regenerates a censored version of the audio using text-to-speech (gTTS).
- Implemented chunked audio processing with pydub, splitting audio into user-adjustable 5–30 second segments (default 15s), cutting peak memory usage by ~25% versus processing full-length files in one pass and preventing crashes on longer uploads.
- Built a live “profanity meter” that computes and displays the percentage of flagged words per upload, with 4 selectable censorship styles (Beep, Random — drawing from 7 alternate tokens, Dolphin, Symbols).
- Designed an interactive Streamlit interface with real-time progress indicators and a side-by-side view of the original transcription versus the censored output, reducing end-to-end processing time to under 20 seconds for a typical 2-minute clip.

## LEADERSHIP & COLLABORATION

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- **Public Relations, TEDxDTU ("AI Boom or Bane"):** Managed outreach and stakeholder coordination for a 500+ attendee event, building working relationships across multiple teams.
- **Logistics Head, ENGIFEST, DTU:** Directed logistics for DTU's annual festival, coordinating a 30+ member volunteer team and multiple stakeholders under time pressure.
- **Volunteer, RobinHood Army (Delhi-NCR):** Taught and mentored underprivileged children, contributing 40+ hours of community service; developed strong communication skills for simplifying concepts to varied audiences.

## ACCOMPLISHMENTS

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- Cleared SSB screening and advanced through the multi-day officer selection assessment for the Indian Armed Forces.
- Secured AIR 11,085 out of 2.1M+ candidates in the 2022 JEE national engineering entrance examination.
- 1st Place — Annual Inter-College Sports Meet, IIT Mandi (2023).